Experiment .: 06

Aim :- To use string functions and regular expressions for counting vowels in a string, data extraction, and text analysis.

Software/Tools Required:
- HTML5 (Markup Language)
- CSS3 (Styling and Layout)
- JavaScript ES6+ (Client-side Interactivity)
- VS Code or any text editor
- Live Server Extension (for local development)

Theory :-

1. String Functions

String functions are built-in JavaScript methods used to perform various operations on strings. Functions such as `length`, `toLowerCase()`, `toUpperCase()`, `charAt()`, `slice()`, and `split()` are useful for processing, modifying, and analyzing textual data.

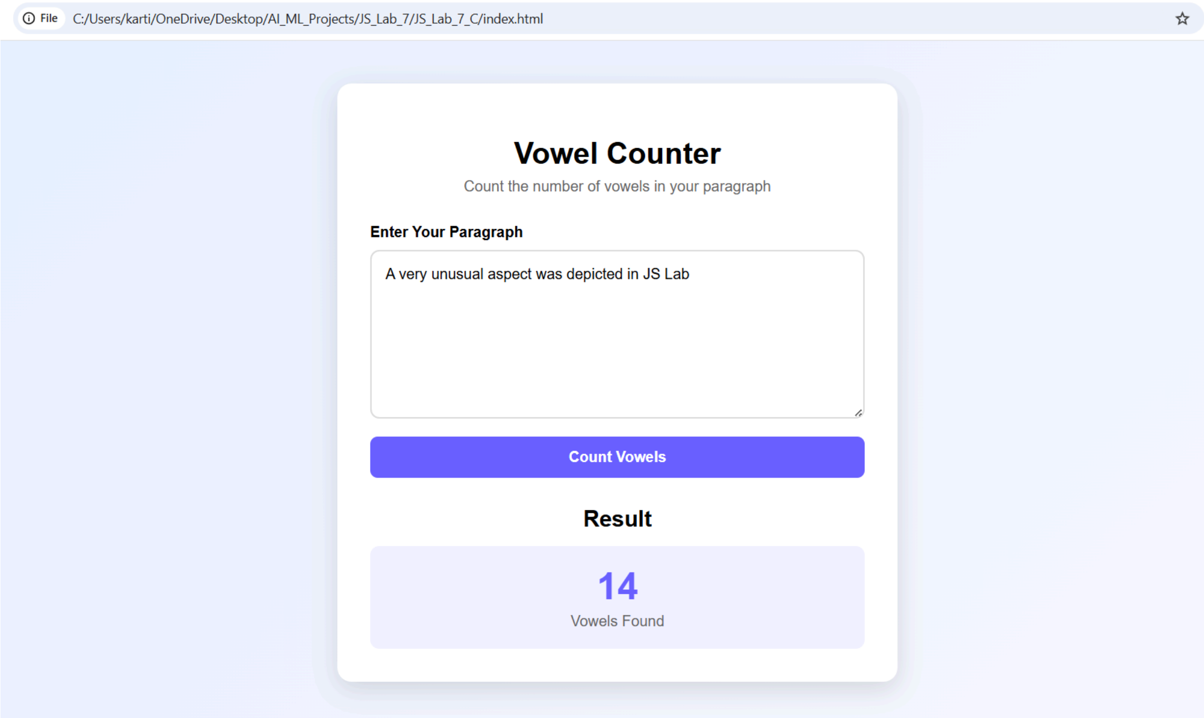
2. Regular Expressions (Regex)

Regular expressions are patterns used to search, match, and extract specific characters or text from a string. In JavaScript, regex can be used with methods such as `match()`, `test()`, and `replace()`. They are particularly useful for identifying vowels, extracting required information, and performing text analysis.

3. Vowel Counting and Data Extraction

Vowel counting involves identifying the occurrences of vowels such as `a`, `e`, `i`, `o`, and `u` within a string. String functions and regular expressions can be combined to count vowels and extract specific patterns or information from textual input efficiently.

Key Features Implemented:
- Text input area for entering strings or paragraphs
- Vowel counting using string functions and regular expressions
- Extraction of specific characters and text patterns
- Basic text analysis using JavaScript
- Dynamic display of analysis results
- CSS Flexbox and responsive styling for better layout


File
C:/Users/karti/OneDrive/Desktop/AI_ML_Projects/JS_Lab_7/JS_Lab_7_C/index.html
Vowel Counter
Count the number of vowels in your paragraph
Enter Your Paragraph
A very unusual aspect was depicted in JS Lab
Count Vowels
Result
14
Vowels Found

Case Study Overview:
This case study demonstrates how to build a functional Text Analysis webpage using vanilla JavaScript, HTML5, and CSS3. The application accepts textual input from the user and processes it using string functions and regular expressions to count vowels, extract relevant data, and perform basic text analysis.

Core Implementation Details:
HTML Structure (index.html):
• Semantic HTML5 elements for better accessibility • Text input section with a textarea and analysis button

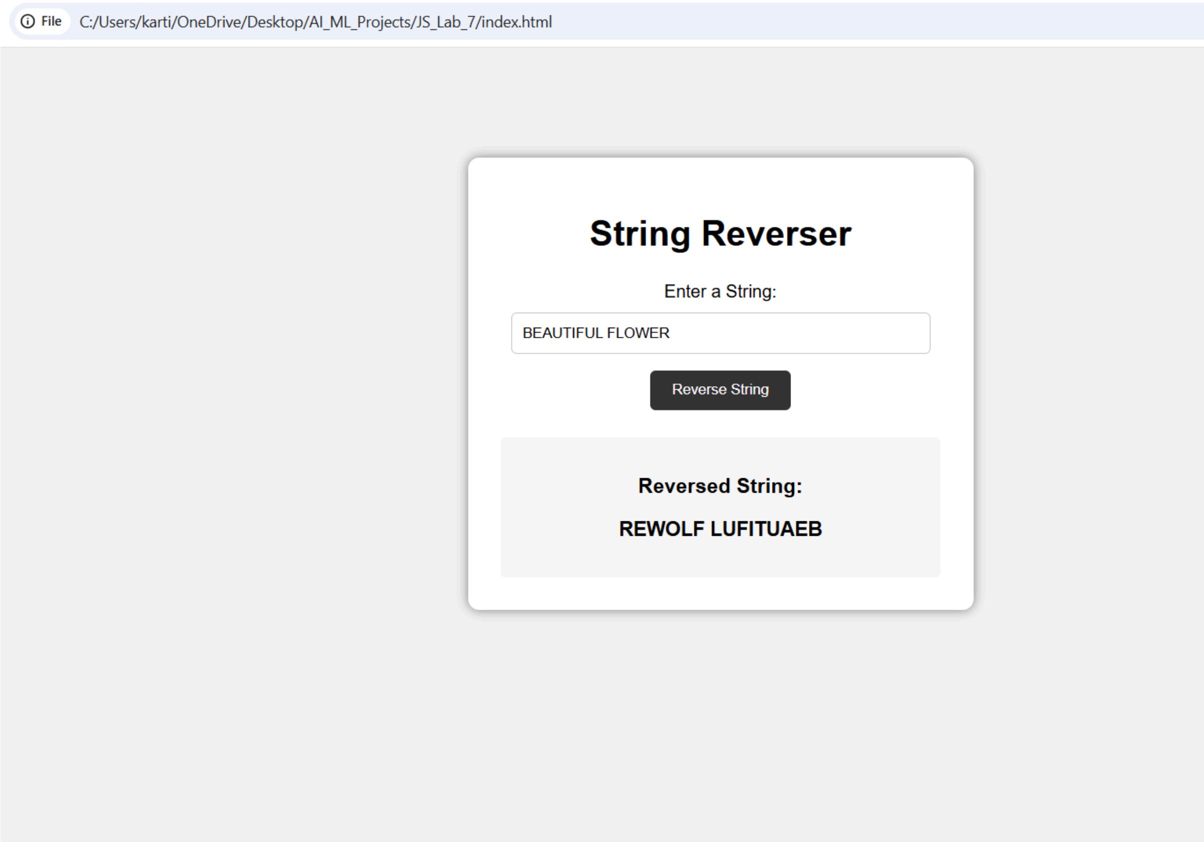
CSS Styling (style.css):
• Responsive container layout for the text analysis interface
• Flexbox for component alignment and spacing

JavaScript Functionality (script.js):
• DOM manipulation to retrieve and display text input • String functions and regular expressions for vowel counting, data extraction, and text analysis

Conclusion:
This Text Analysis application successfully demonstrates essential web development concepts by combining HTML5 for structure, CSS3 for styling and layout, and JavaScript for text processing and interactivity. The project demonstrates practical skills in:

- JavaScript string functions and string manipulation
- Regular expressions for pattern matching and vowel identification
- Data extraction and basic text analysis techniques
- Client-side JavaScript programming and DOM manipulation
- User input handling and dynamic result display
- Processing textual data efficiently using built-in JavaScript features

The application serves as a foundation for more advanced text-processing applications and demonstrates the ability to analyze, extract, and manipulate textual data using JavaScript and regular expressions without external frameworks.


File
C:/Users/karti/OneDrive/Desktop/AI_ML_Projects/JS_Lab_7/index.html
String Reverser
Enter a String:
BEAUTIFUL FLOWER
Reverse String
Reversed String:
REWOLF LUFITUAEB